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RESEARCH ARTICLE

SAFF-ConvNeXt: A Scale-Aware Feature Fusion and SE-Attention Enhanced ConvNeXt Framework for Kidney Disease Detection in Medical Imaging

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ABSTRACT Chronic Kidney Disease (CKD) is a serious global health issue that affects more than 10% of the population, often undetected until advanced stages. CKD can lead to severe complications, including kidney failure, anemia, osteoporosis, and heart disease. Early detection is critical for improving patient outcomes, yet current diagnostic methods lack early, non-invasive solutions. This study proposes a novel deep learning (DL) architecture for the early detection of Chronic Kidney Disease (CKD), using a pretrained ConvNeXtBase backbone enhanced by multi-scale feature fusion and Squeeze-and-Excitation (SE) attention mechanisms. Unlike traditional single-path CNNs, the SAFF-ConvNeXt model utilizes a ConvNeXt-based architecture with embedded SE attention to enhance feature representation. It implements a multi-scale fusion strategy that captures both fine-grained and high-level semantic features effectively. The Grad-CAM technique was integrated to enhance interpretability by providing visual explanations of model predictions. Extensive experiments were conducted using the CT kidney (CTK) dataset, complemented by ablation studies to identify optimal hyperparameters. Comprehensive statistical analysis and comparisons with five DL models—DenseNet121, EfficientNet-B0, MobileNet, VGG19, and Xception—demonstrated the superiority of the proposed model. It achieved exceptional performance metrics, including 99.75% accuracy, 99.83% specificity, 99.38% sensitivity, 99.42% precision, and 99.39% F1-score. This approach is widely used in medical image analysis to lessen the workload of radiologists, improve clinical decision-making, and enhance patient education regarding the management of CKD progression.

INDEX TERMS Kidney, stones, cysts, tumors, deep learning, ConvNeXtBase, CT scans, Grad-CAM.

I. INTRODUCTION

Chronic kidney disease (CKD), often caused by diabetes, hypertension, infections, or genetic factors, progresses through five stages based on declining glomerular filtration rate (GFR), with advanced stages (3–5) posing high cardiovascular and mortality risks. Over 800 million people globally are affected, disproportionately impacting older adults, diabetics, and socio-economically disadvantaged groups. Acute kidney injury (AKI) arises suddenly from trauma or toxins [1], [2], [3]. CKD is frequently asymptomatic early on (''silent killer''), with later symptoms including hematuria, nausea, breathing difficulties, anemia, and electrolyte imbalances [4], [5].

Complications include kidney stones (mineral deposits causing severe pain and urinary issues) [6], cysts (simple vs. complex; complex cysts may indicate polycystic kidney disease or tumors) [7], and tumors like renal cell carcinoma (RCC) [8].

Diagnosis involves GFR, serum creatinine, imaging (ultrasound, MRI, CT), and biopsy. Treatment ranges from lifestyle modifications and managing comorbidities

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to dialysis/transplant for kidney failure [9], [10]. Rising CKD prevalence, coupled with healthcare resource shortages (especially in developing regions), highlights the critical need for cost-effective diagnostic models to aid clinicians in early detection and decision-making [11].

Artificial Intelligence (AI) is a rapidly evolving field that focuses on creating machines capable of human-like tasks. In healthcare, AI has become a transformative tool in classifying CKD by analyzing patient records, lab results, and demographic data to identify risk factors [12], [13]. Moreover, AI helps to overcome challenges such as diagnostic delays, resource limitations, and human error. Machine learning (ML) and Deep Learning (DL) models analyze various types of data, including imaging (ultrasound, MRI, CT scans), laboratory results (serum creatinine, GFR), and electronic health records. These analyses help detect patterns that indicate conditions like kidney stones, cysts, and tumors [14]. For example, AI algorithms can distinguish between simple and complex kidney cysts on imaging, identify early-stage renal cell carcinoma (RCC) by recognizing subtle tumor features [7], [11], [15]. CNNs are particularly effective at processing radiological images, allowing for automated segmentation of kidney structures and detection of anomalies.

Computed tomography (CT) provides high-resolution, three-dimensional anatomical details essential for detecting renal anomalies. However, the global shortage of nephrologists and radiologists—especially acute in regions like South Asia, with fewer than one nephrologist per million people—limits timely and accurate diagnosis. In this context, artificial intelligence (AI), especially convolutional neural networks (CNNs) based DL, offers a promising solution for augmenting diagnostic capabilities. These models excel at processing radiological images and can be trained to detect kidney stones, cysts, and tumors automatically. Despite some progress in this domain, existing studies are limited by small datasets and often focus on a single condition rather than multi-class classification. Moreover, most prior work relies on traditional machine learning methods and ultrasound data, which are less comprehensive than CT scans [16].

Given the burden of kidney diseases, the limited availability of specialists, and the advancement in AI-based vision systems, there is an urgent need to develop robust, automated diagnostic models capable of detecting multiple kidney conditions from radiological data. Such tools can assist clinicians in early detection, reduce diagnostic workload, and improve patient outcomes—particularly in resource-constrained settings [16].

The early detection of CkD is critical for improving patient outcomes, yet current diagnostic methods lack early, non-invasive solutions. Hence, this research proposes a new DL architecture for detecting kidney disease that utilizes a pre-trained ConvNeXtBase backbone, enhanced by a multi-scale feature fusion strategy and attention mechanisms. Unlike traditional single-path CNN classifiers, our model extracts and integrates features from early, intermediate, and deep stages of the ConvNeXt hierarchy. This approach allows it to capture both detailed local textures and high-level semantic patterns relevant to pathological structures. To highlight diagnostically significant regions, we incorporate lightweight SE blocks at each scale, enabling the model to adjust the importance of spatial and channel features dynamically. The proposed integration enhances feature representation and model efficiency, enabling improved performance while maintaining low computational complexity. Moreover, the proposed architecture shows improved classification performance on multi-class CKD datasets and offers interpretable feature activation through its hierarchical fusion design. To our knowledge, this is the first work that combines ConvNeXt with scale-aware fusion and channel attention specifically for kidney disease detection in medical imaging. The ConvNeXtBase model [17] consists of convolutional layers, normalization, and other architectural improvements aimed at enhancing performance on visual tasks. It uses modern tweaks to traditional CNNs, such as large kernel sizes, fewer activation functions, and layer normalization.

The Grad-CAM method is used to pinpoint areas of vascular thickening and highlight the widespread distribution of opacities in scanned images. The doctors can use this technique to realize important features related to CKDs.

Our DL model improves the early detection of CKDs by using the CTK dataset, which comprises four different categories: cysts, stones, normal, and tumors. The first steps in preparing the CTK dataset included resizing and normalization. The dataset was then split, with 85% allocated for training (10,579 CT scans) and 15% for testing (1,867 CT scans). The undersampling on the training dataset is carried out. Testing was conducted using the testing set of the CTK dataset, and comparisons with existing models revealed that our model surpasses other DL. The motivation for this study arises from the need for a lightweight yet highly accurate model suitable for CKD detection. Below the contributions from our research are summarized:

1) SAFF-ConvNeXt, a new hybrid DL architecture, is introduced for detecting CKDs, combining a pretrained ConvNeXtBase backbone with a custom classification head. The model incorporates multi-scale feature fusion by extracting and combining representations from the early, middle, and late stages of the ConvNeXt model. This approach captures both low-level and high-level semantic information important for understanding kidney pathology. SE attention modules are implemented at each selected scale to enhance channel-wise feature representations and focus on disease-relevant features within the fused context.
2) The design is lightweight and modular, ensuring that the architecture remains computationally efficient while achieving higher accuracy, making it suitable for clinical use or deployment on systems with limited resources.
3) Ablation study was performed to find optimal hyper-parameters. Moreover, an extensive experiment was

conducted using the CTK dataset, which included the application of Grad-CAM to interpret model decisions and provide visual explanations.

4) Full depth statistical analysis was performed among all methods.
5) Compared to five DL models (Densent121, EfficientNet-B0, MobileNet, VGG19, and Xception), the proposed model demonstrated outstanding effectiveness in accurately managing multi-classification tasks. The SAFF-ConvNeXt model achieved impressive outcomes, with an average accuracy of 99.75%, specificity of 99.83%, sensitivity of 99.38%, precision of 99.42%, and an exceptional F1-score of 99.39%.

The study is divided into multiple components. A review of the most recent CKD diagnosis systems reported in previous studies is provided in Section II. The preprocessing procedures for the CTK dataset are described in Section III, along with the technique used for the suggested model. In Section IV, the SAFF-ConvNeXt model's results are examined. Finally, the implications drawn from this study are summarized in Section V.

II. PREVIOUS STUDY

The diagnosis of CKD is a widely studied topic in the field of medical image analysis. Many research projects investigate this matter from diverse perspectives. For example, Hossain et al. [18] worked on enhancing the quality of CT scan images by using techniques such as cropping, resizing, and contrast limited adaptive histogram equalization (CALHE). The CTK dataset was split with 80% allocated for training purposes and 20% set aside for testing in all instances. After improving the images, they extracted features using a new method called adaptive local binary pattern (ALBP), which they compared to the advanced local binary pattern (LBP) method. These features were then used with classifiers including Random Forest (RF), Decision Tree (DT), Naive Bayes (NB), K-Nearest Neighbor (KNN), and Support Vector Machine (SVM). Additionally, they explored an ensemble model that used soft voting to create a more reliable overall model. By applying their feature descriptor and combining five techniques (SVM, DT, RF, KNN, and NB) through the voting approach, they attained an accuracy exceeding 99%.

Islam et al. [16] collected and analyzed a total of 12,446 complete urogram and abdominal CT images, which included categories for normal, tumor, stone, and cyst. An exploratory data analysis of these images showed that all categories had a similar distribution of mean variation. The authors introduced six models for analysis: three based on vision transformers (SWIN, CCT, and EANet) and three utilizing popular DL models (Inceptionv3, ResNet, and VGG16). Among these models, the SWIN transformer exhibited the highest accuracy at 99.30%. Furthermore, performance measures such as F1-score, precision, and recall indicated that the SWIN transformer outperformed all other models. The findings also revealed that VGG16 performed better than Inceptionv3 and ResNet50 in detecting significant anatomical abnormalities. The findings from these models were subsequently clarified using the Grad-CAM explainable AI technique.

Hossain et al. [19] created a convolutional neural network (CNN) to categorize CT scan images into specific groups. They utilized a custom CNN model along with popular pre-trained models such as ResNet50, VGG19, InceptionV3, and SqueezeNet. The models were trained on the CTK dataset, which contained CT images of both healthy and diseased kidneys. The process consisted of two main steps: first, image segmentation was performed using the watershed algorithm to identify the kidney areas in the CT scans. Then, these segmented images were analyzed and classified by the DL models. The results indicated that the VGG19 model got the maximum accuracy at 99.9%, with InceptionV3 closely following at 98.8%.

Afroj et al. [20] introduced a new DL method for accurately classifying CT images of the kidneys. Their approach aimed to automate the classification of kidney conditions, specifically focusing on distinguishing between Normal, Cyst, Tumor, and Stone categories. The CNN enhanced with attention layers utilizes its capability to capture complex patterns and features in the images, allowing for precise identification and differentiation of kidney abnormalities. Experimental results from the CTK dataset demonstrated an accuracy of 97.98%, along with an average precision, detection, and F1 score of 98%.

Majid et al. [21] investigated the use of ML and fine-tuned Transfer Learning (TL) to improve the detection of kidney tumors (KT). They trained models using CT images from patients both with and without kidney tumors. The study explored three different image resolutions: 32×32, 64×64, and 128×128 pixels, and utilized the Grey Level Co-occurrence Matrix (GLCM) for feature extraction. The GLCM analyzed how often pixel pairs appeared based on their distance (d) and angle (θ) within the images. Various detection techniques, including RF, SVM, Gradient Boosting (GB), and Light Gradient Boosting Model (LGBM), were used to identify kidney tumors in CT images for diagnostic purposes. Additionally, the research included tests with fine-tuned versions of the ResNet-101 and DenseNet-121 models to enhance computer-assisted diagnosis of kidney tumors. The effectiveness of these fine-tuned models was evaluated by comparing their performance with the four ML models (RF, SVM, LGBM, and GB). Notably, ResNet-101 and DenseNet-121 achieved the highest accuracy of 94.09%, precision of 95.10%, recall of 93.5%, and F1-score of 93.95% when analyzing input images at a resolution of 32×32 pixels.

Pande and Agarwal [22] developed a YOLOv8 DL model to aid in the identification of kidney diseases, addressing the worldwide shortage of urologists. They established a new and adaptable platform for the clinical assessment of kidney problems such as tumors, stones, and cysts. The YOLOv8 model achieved an accuracy of 82.52%, a precision

of 85.76%, a recall of 75.28%, an F1 score of 75.72%, and a specificity of 93.12%.

Islam et al. [23] developed a DL framework incorporating spatial attention. Spatial attention modules were smoothly integrated into the architecture of CNNs to enhance interpretability, providing insights into key areas that influence the model's decisions. This improvement strengthened the model's ability to process visual data effectively, resulting in better classification performance. The proposed CNN model guided by spatial attention demonstrated an impressive accuracy of 99.52%, along with precision, recall, and f1-score metrics of 99.54%, 99.52%, and 99.52% respectively.

Yildirim et al. [24] introduced a new method for detecting stones using CT images of coronal in combination with cutting-edge DL technology. This development represented a significant advancement in AI. The proposed system processed 1,799 images from different cross-sectional CT scans of patients and achieving an accuracy of 96.82% in identifying stones of kidney. The new method was effective in accurately detecting even small kidney stones. Furthermore, when applied to a larger dataset consisting of 433 individuals, the cutting-edge DL model yielded outstanding results, showcasing its potential for real-world clinical.

Qadir and Abd [25] introduced a technique that integrated the DenseNet-201 DL model, which was a pre-trained DL model used for extracting features, with RF, a ML method, to identify CKDs, including stones, cysts, tumors, and normal tissue. This combined approach was evaluated on the CTK dataset. The CTK dataset was divided into 80% allocated for training and 20% reserved for testing. This hybrid model reached an accuracy of 99.44%.

Almuayqil et al. [26] developed KidneyNet, a sophisticated computer-aided diagnosis (CAD) system to detect CKDs, including kidney tumors, cysts, and stones in CT scans. KidneyNet was built on a CNN architecture comprising two fully connected layers, eight convolutional layers, a flattening layer, and three pooling layers. The model utilized the gradient-weighted class activation mapping (Grad-CAM) technique to highlight the affected regions in the scans, enhancing interpretability by enabling clinicians to visualize the areas considered significant for identifying abnormalities such as tumors, cysts, or stones. Upon thorough evaluation using a CT kidney dataset, KidneyNet achieved outstanding performance, reaching an accuracy of 99.88%.

The limitations of the previous studies are as follows:

1) To our knowledge, this is the first study to integrate ConvNeXt with scale-aware feature fusion and SE attention mechanisms specifically designed for CKDs detection in medical imaging.
2) The authors of the aforementioned research focused on metrics like accuracy, precision, recall, and F1 score; however, they did not evaluate results using statistical confidence intervals (CIs), margin of error, and the Friedman ranking with the Nemenyi post-hoc test. In contrast, we have successfully employed statistical methods to analyze and compare outcomes from various DL approaches by investigating CIs and utilizing the Friedman ranking along with the Nemenyi post-hoc test. This highlights the distinctive benefit of our methodology.
3) The authors of the previously mentioned research did not conduct an ablation study. In contrast, we conducted this research to understand how each component or feature of our proposed model affects performance by systematically altering or removing them and analyzing the resulting impact.

TABLE 1. The allocation of CT images within the CTK dataset.

Class	Image Count
Cyst	3709
Normal	5077
Stone	1377
Tumor	2283
Total	**12,446**

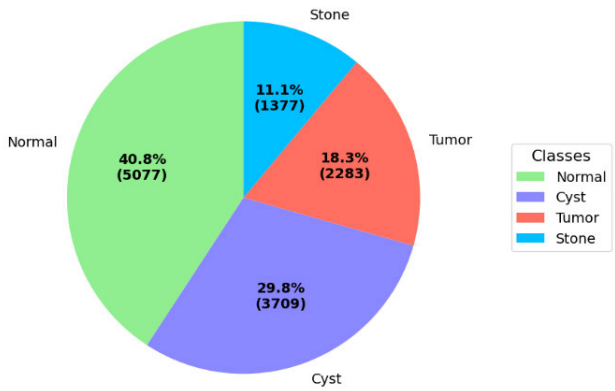


FIGURE 1. The distribution of the CTK's CT images.

III. DATASET AND METHOD

A. CTK DATASET

Islam et al. [16] developed the CTK dataset form various hospitals in Bangladesh. The CTK dataset is accessible through the Kaggle environment [27]. It comprised images of various patients exhibiting healthy kidneys, as well as those with stones, tumors, or cysts. The dataset includes both coronal and axial views from non-contrast and contrast urogram and abdominal CT scans, totaling 12,446 images: 5,077 normal CT images, 3,709 cyst CT images, 1,377 stone CT images, and 2,283 tumor CT images, as outlined in Table 1. As shown in Table 1 and Figure 1, the CTK dataset exhibits an imbalance. Each CT image has a resolution of 512 × 512 pixels. Initially, the CT images were in Digital Imaging and Communications in Medicine (DICOM) format and were changed to JPEG format using the Sante DICOM tool. The images were annotated with the help of the Philips IntelliSpace Portal application and underwent review by technicians and medical professionals to ensure accuracy.

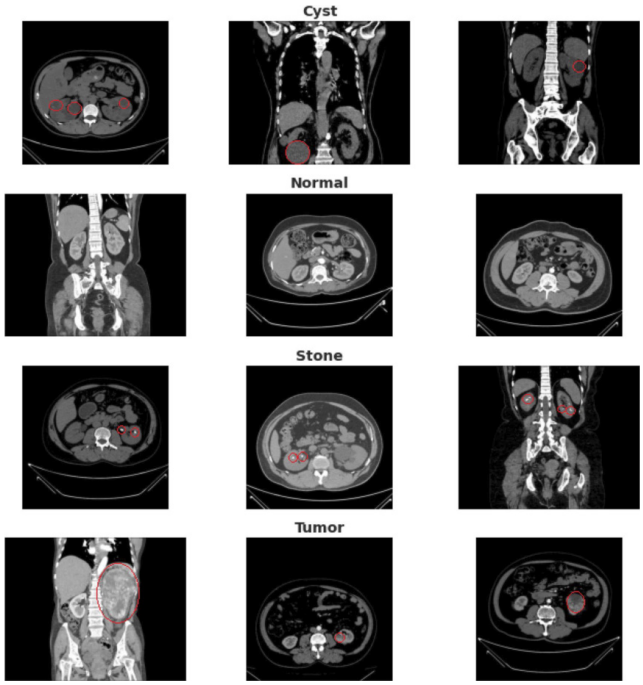


FIGURE 2. Examples of CT scans from the CTK dataset.

Figure 2 displays samples of CT images from the CTK dataset.

B. METHODOLOGY

We introduced SAFF-ConvNeXt, an innovative DL framework for detecting CKDs, utilizing a pretrained ConvNeXtBase backbone enhanced by a strategy for multi-scale feature fusion and attention mechanisms. Unlike conventional single-path CNN classifiers, our model extracts and integrates features from the initial, intermediate, and deeper levels of the ConvNeXt hierarchy. This approach captures both detailed local textures and high-level semantic patterns that are significant for identifying pathological structures. To further highlight diagnostically relevant areas, we incorporate lightweight SE blocks at each scale, allowing the model to dynamically adjust the importance of spatial channels. This architecture demonstrates enhanced classification accuracy on multi-class kidney disease datasets and offers interpretable feature activation through its hierarchical fusion design.

The proposed model improves the early detection of CKDs for healthcare providers by effectively identifying small changes in imaging data that serve as early indicators of the disease. By recognizing these changes at an early stage, clinicians can act before the disease worsens. Its high accuracy, adaptability, and ability to analyze complex patterns allow for earlier diagnoses, which are essential for slowing disease progression, improving patient outcomes, and reducing the time and costs related to diagnosis. To assess the effectiveness of this approach, an experiment was performed utilizing the CTK dataset, which comprises four categories: normal, stones, tumor, and cyst. The preliminary processing of the CTK dataset involved resizing and normalization methods. Following this, the dataset was partitioned into 85% for training (10,579 CT images) and 15% for testing (1,867 CT images).

The effectiveness of the proposed classifier was evaluated in relation to five DL models: DenseNet121, EfficientNet-B0, MobileNet, VGG19, and Xception. The work flow steps are illustrated in Figure 3. The steps for training and implementing the proposed model, as well as the five other DL models, are outlined in Algorithm 1. The procedure for the SAFF-ConvNeXt model is as follows:

1) **Stage 1 (Dataset Normalization and Resizing)**: In this stage, the CT scans were normalized and resized to ensure consistency in their dimensions and appearance.
2) **Stage 2 (Dataset Splitting)**: The CTK dataset was split into training and testing sets. The training set comprised 85% of the data, consisting of 10,579 images, and the testing set made up the remaining 15%, totaling 1,867 instances.
3) **Stage 3 (Undersampling)**: In stage 3, we utilized the undersampling technique to balance the training set, decreasing the total number of CT images from 10,579 to 2,000. This resulted in each of the four classes containing 500 CT images.
4) **Stage 4 (The Pre-Training Six DL Models)**:During stage 4, training the SAFF-ConvNeXt, DenseNet121, EfficientNet-B0, MobileNet, VGG19, and Xception models using the ImageNet dataset were conducted.
5) **Stage 5 (5-Fold Cross Validation)**: During this stage, the training data was divided into five equal parts. One of the five partitions was utilized as the validation set, while the remaining four partitions were employed as the training set. Each model was trained on the training set and then evaluated or tested on the validation set. To make sure that every partition was used as the validation set precisely once, this procedure was carried out five times.
6) **Stage 6 (Calculating the Measured Metrics)**: During this stage, following the completion of five iterations, we computed the mean values of the measured metrics, including F1-score, precision, FNR, accuracy, recall, and NPV across the five folds.
7) **Stage 7 (Multi-classification)**: In this stage, the six models were tested on the testing data from the CTK dataset.

1) DATA PRE-PROCESSING

We used three pre-processing techniques to process our dataset before performing actual experiments. First, resizing and normalization techniques were performed before splitting the dataset into training and test. Then, undersampling technique is carried out on the training dataset. Below, the techniques are described:

1) **Resizing:** Ensures all images have the same dimensions as required by the model.

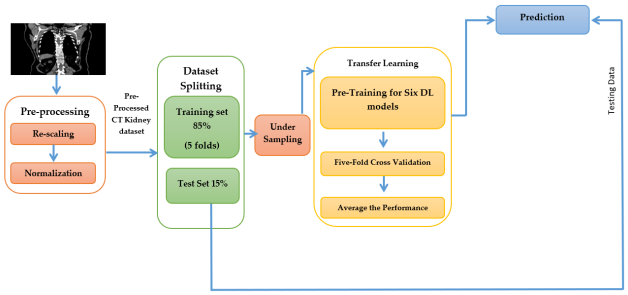


FIGURE 3. The work flow steps.

2) **Normalization:** Scales the pixels to a range, typically [0, 1], using min-max scaling:

$$x = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{1}$$

3) **Undersampling:** Decreases the number of instances in one or more categories within the dataset. It is usually implemented to tackle the issue of class imbalance, which occurs when one or more classes contain a significantly larger number of samples compared to others.

In our methodology, images from the CTK dataset were resized to 224 × 224 pixels. Then, images were transformed from a range of 0 to 255 to a normalized scale of 0 to 1 to maintain consistency. After that, we split the dataset into training (85%) and test (15%).

We also implemented an undersampling technique to balance the training set, reducing the total from 10,579 to 2,000 CT images and ensuring that each of the four classes contained 500 images. In undersampling, a controlled strategy capped each class at 500 images. For classes with more than 500 images, a random subset of 500 samples was retained, while smaller classes were kept in full. This ensured uniform representation and reduced bias from overrepresented categories. The choice of undersampling was made for practical reasons, including enhanced computational efficiency (by decreasing the dataset size for faster training) and minimizing the risks of overfitting associated with synthetic oversampling methods, which could result in unrealistic data patterns. Table 2 illustrates the distribution of the CTK dataset after applying the undersampling technique.

2) ConvNeXtBase MODEL

ConvNeXtBase is a CNN architecture designed to enhance the efficiency and performance of traditional CNNs for image recognition tasks. Developed by the Facebook AI (Meta) research team, ConvNeXtBase borrows ideas from Vision Transformers (ViTs) to improve the capabilities of CNNs while retaining their core benefits, such as inductive biases and efficiency. The architecture of ConvNeXtBase shown in Figures 4 and 5 is built upon ResNet-style models, with several key adjustments:

Algorithm 1 The Training and Evaluation of the Six DL Model

Input: CT kidney dataset
Output: Six DL model for CKDs

BEGIN
  STEP 1: Resizing and Normalization of images
    FOR EACH image **IN** the *CTK* **DO**
      Resize an image pixel to 224 × 224.
      Normalize image pixel values from [0, 255] to [0, 1].
    END FOR
  STEP 2: Dataset Splitting
    PARTITION *CTK* **INTO**
      Training set → 85%.
      Testing set → 15%.
  STEP 3: UNDERSAMPLING DATASET
    UNDERSAMPLING THE TRAINING SET FROM 10,579 TO 2,000.
  STEP 4: Six DL Training
    FOR EACH DL **IN** [SAFF-ConvNeXt, DenseNet121, EfficientNet-B0, MobileNet, VGG19, and Xception] **DO**
      Load and pre-train DL on the ImageNet dataset.
    END FOR
  STEP 5: Five Fold Cross Validation
    FOR EACH pre-trained DL **IN** [SAFF-ConvNeXt, DenseNet121, EfficientNet-B0, MobileNet, VGG19, and Xception] **DO**
      FOR EACH fold k = 1 to 5
        Use the other 4 folds (not including fold k) as the training data.
        Use fold k as the validation data.
        Evaluate the trained model on the validation set (fold k) and record the performance metrics.
      END FOR
    END FOR
  STEP 6: Average the Performance Metrics
    FOR EACH DL model **IN** [SAFF-ConvNeXt, DenseNet121, EfficientNet-B0, MobileNet, VGG19, and Xception] **DO**
      Calculate the average of the performance metrics for DL.
    END FOR
  STEP 7: Multi-classification
    FOR EACH DL model **IN** [SAFF-ConvNeXt, DenseNet121, EfficientNet-B0, MobileNet, VGG19, and Xception] **DO**
      Assess the DL model on the testing set.
    END FOR
END

1. **Large Kernel Convolutions**: The model utilizes larger 7 × 7 convolutional kernels, inspired by the approach used in ViTs.

TABLE 2. The distribution of the CT images in the CTK dataset after applying the undersampling technique.

Class	Image Count
Cyst	500
Normal	500
Stone	500
Tumor	500
Total	**2000**

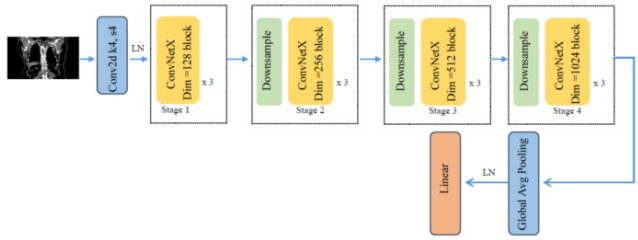


FIGURE 4. ConvNeXtBase's architecture.

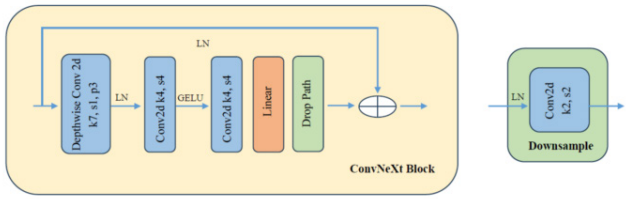


FIGURE 5. ConvNeXt block's architecture.

2. Layer Normalization: Instead of batch normalization, ConvNeXtBase employs layer normalization to improve the model's performance.

3. Reduced Activation Functions: The number of activation functions within the network has been reduced to streamline the overall design.

4. Scalable Network Design: The architecture has been optimized for scalability and enhanced performance.

ConvNeXtBase is part of the broader ConvNeXt family of models, which includes versions with different scales, such as ConvNeXt-Tiny, ConvNeXt-Large, and so on, as described in [20]. The advantages of ConvNeXtBase are as follows:

1. Exceptional Performance: ConvNeXtBase demonstrates competitive accuracy in image classification and other visual tasks, matching the performance of modern Vision Transformers (ViTs) on benchmark datasets like ImageNet.

2. Computational Efficiency: Despite its competitive performance, ConvNeXtBase retains the simplicity and computational efficiency of traditional CNNs, making it more resource-efficient than transformer models in certain scenarios.

3. Scalable Design: ConvNeXtBase is designed to be scalable across different model sizes, from lightweight versions like ConvNeXt-Tiny to larger models like ConvNeXt-Large, providing flexibility depending on computational requirements.

4. Enhanced Convolutional Architecture: By introducing larger kernel sizes, layer normalization, and simplified activation patterns, ConvNeXtBase improves upon traditional convolutional layers without sacrificing their benefits, such as spatial locality and translation invariance.

5. Stable Training: ConvNeXtBase introduces structural improvements that stabilize the training process and lead to better generalization on a wide range of datasets.

While ConvNeXtBase maintains the fundamental structure of convolutional layers, its mathematical model can be divided into several critical components, each of which helps the network learn and make predictions. The ConvNeXtBase's mathematical model can be broken down as follows [28]:

1) **Convolutional Layer:** The fundamental operation in ConvNeXt is the convolution operation, which applies a kernel (filter) to the input to extract features. Mathematically, the convolution of an input X with a filter W (of size $k \times k$) at a position (i, j) is given by:

$$Y(i,j) = \sum_{m=1}^{k} \sum_{n=1}^{k} W(m,n) \cdot X(i+m-1, j+n-1) \tag{2}$$

where: X is the input feature map, W is the convolution kernel (weights), and $Y(i, j)$ is the output feature map. ConvNeXt uses large convolution kernels (e.g., 7×7) to capture more context.

2) **Layer Normalization:** Instead of traditional batch normalization, ConvNeXtBase uses Layer Normalization to normalize activations across each layer. This ensures smoother training and better generalization. For a given input x, layer normalization is defined as:

$$\hat{x} = \frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}} \tag{3}$$

where μ is the mean of the layer activations, σ^2 is the variance, and ϵ is a minor constant used to ensure stability.

3) **Activation Function (GELU):** ConvNeXtBase uses the GELU (Gaussian Error Linear Unit) activation function, which is defined as:

$$\text{GELU}(x) = x \cdot \Phi(x) \tag{4}$$

where $\Phi(x)$ is the cumulative distribution function of the Gaussian distribution:

$$\Phi(x) = \frac{1}{2}\left[1 + erf\left(\frac{x}{\sqrt{2}}\right)\right] \tag{5}$$

where $erf\left(\frac{x}{\sqrt{2}}\right)$ is the error function, is a special mathematical function used in probability, statistics, and partial differential equations. It represents the probability that a normally distributed random variable falls within a certain range. This allows for smooth,

differentiable activation that can better approximate nonlinear functions.

4) **Downsampling Layers (Patchify):** Instead of typical max-pooling or strided convolutions, ConvNeXtBase uses downsampling layers (like in ViTs). These downsampling layers reduce the spatial dimension by reshaping the input into patches. The mathematical formulation for downsampling is:

$$Y = \text{Reshape}(X) \tag{6}$$

where X is the input, and the function reshapes it into patches, reducing the spatial dimensions and increasing feature depth.

5) **Residual Connections:** Residual connections allow for better gradient flow in deep networks. Mathematically, the residual block is defined as:

$$Y = X + f(X) \tag{7}$$

where X is the input, $f(X)$ is the transformation (e.g., convolution, normalization, activation), and the addition represents the residual connection, allowing the network to learn identity mappings.

6) **Overall Network Forward Pass:** A forward pass through ConvNeXtBase can be expressed as the composition of multiple operations (convolution, layer normalization, activation, downsampling, and residuals):

$$Y = ResidualBlock_n \circ \cdots \circ ResidualBlock_1 \quad (X) \tag{8}$$

where:

- Each residual block consists of convolution, normalization, and activation functions.
- The composition ($\circ$) refers to the successive application of each block.
- X is the input scan or feature map, and Y is the final output after processing through all layers.

7) **Objective Function (Cross-Entropy Loss):** During training, ConvNeXtBase typically uses cross-entropy loss for classification tasks. For a given prediction p and true label y, the cross-entropy loss is defined as:

$$L(p, y) = -\sum_{i=1}^{C} y_i \log(p_i) \tag{9}$$

where C is the number of classes, p_i is the predicted probability for class i, and y_i is the true label.

3) SAFF-ConvNeXt MODEL

The proposed SAFF-ConvNeXt model is an integration of the ConvNeXtBase model, SE blocks as attention modules, and a multi-scale feature fusion approach. Its architecture is shown in Figure 6. In SAFF-ConvNeXt model, the ConvNeXtBase model was utilized as the main feature extractor because of its hierarchical convolutional design and enhanced efficiency compared to traditional CNN architectures. The input image

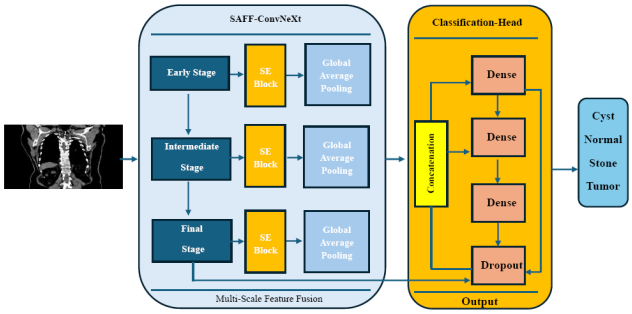


FIGURE 6. SAFF-ConvNeXt's architecture.

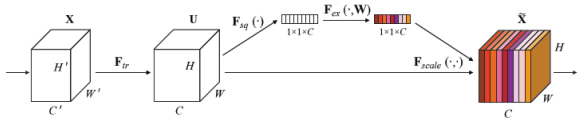


FIGURE 7. A Squeeze-and-Excitation block [29].

was processed through various stages of the ConvNeXtBase, generating multi-level feature maps that captured both low-level spatial details and high-level semantic content.

To enhance the channel interdependencies within each feature map, SE attention blocks were integrated after specific stages of the ConvNeXtBase. Each SE block executes a two-step process:

1) Squeeze: This step involves applying global average pooling across the spatial dimensions to create a channel descriptor that encapsulates global contextual information.
2) Excitation: The descriptor is then fed into two fully connected layers followed by a sigmoid activation function to produce adaptive channel weights. These weights are applied to the feature maps through channel-wise multiplication, which highlights the informative channels while diminishing the less relevant ones.

The multi-scale feature fusion module merged feature maps from various ConvNeXtBase stages, ensuring that both spatial granularity and semantic richness were maintained. The outputs from multiple scales were first adjusted to a common spatial resolution using bilinear interpolation or convolutional projection. These normalized feature maps were then either concatenated or combined element-wise, followed by a convolutional refinement layer to effectively integrate the fused features.

Unlike conventional single-path CNN classifiers, our system extracts and integrates features from early, intermediate, and deep levels of the ConvNeXt architecture, effectively capturing both detailed local textures and higher-level semantic patterns that are important for identifying pathological structures. To highlight diagnostically relevant areas, we incorporate lightweight SE blocks at every scale, allowing

the model to adaptively adjust the importance of spatial and channel features. This architecture shows superior classification performance on multi-class CTK dataset and offers interpretable feature activation through its hierarchical fusion design.

The SE block as shown in Figure 7[1] is a mechanism for attention that aims to enhance the representational capabilities of CNNs by explicitly addressing the relationships between the channels of feature maps. This concept was presented by Hu et al. [30] in their paper "Squeeze-and-Excitation Networks". It has gained significant popularity in medical imaging applications due to its effectiveness in emphasizing features that are important for disease detection. The SE block consists of three key operations:

- Module 1: Squeeze – Global Information Embedding:
  - **Input:** A 3D feature map of shape $H \times W \times C$, where H and W are spatial dimensions and C is the number of channels.
  - **Operation:**
    - Applies Global Average Pooling (GAP) across the spatial dimensions.
    - This reduces each channel to a single scalar, summarizing the global spatial context.
  - **Output:** A 1D vector of size C, representing channel-wise descriptors. For an input tensor $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, this produces $\mathbf{z} \in \mathbb{R}^C$ by computing:

$$z_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} X_{i,j,c}$$

  - **Purpose:** To compress spatial information into a compact form that retains the essence of each channel.
- Module 2: Excitation – Channel-Wise Attention Learning
  - **Input:** The C-dimensional vector from the squeeze module.
  - **Operation:**
    - Passes the vector through a two-layer Fully Connected (FC) bottleneck:
      1) First FC layer: Reduces dimension to C/r using ReLU activation.
      2) Second FC layer: Expands back to C using sigmoid activation.
    - The result is a set of attention weights $\mathbf{s} \in [0, 1]^C$.
  - **Output:** A weight vector of size C, where each element reflects the importance of a feature channel.
  - **Purpose:** To learn which channels are most relevant for the task and allow the model to adaptively emphasize them.
- Module 3: Recalibration – Feature Reweighting
  - **Input:** Original feature map $\mathbf{X}$ and learned weights $\mathbf{s}$.
  - **Operation:**
    - Each channel $\mathbf{X}_c$ in the original feature map is multiplied by its corresponding weight s_c:

$$\tilde{\mathbf{X}}_c = s_c \cdot \mathbf{X}_c \quad \text{for } c = 1, \ldots, C$$

  - **Output:** A recalibrated feature map with the same shape as the input, $H \times W \times C$.
  - **Purpose:** To enhance important features and suppress less useful ones, improving the model's focus and performance.

The following presents a comprehensive modular and mathematical analysis of the proposed SAFF-ConvNeXt model for detecting CKDs. The model utilizes multi-scale feature integration and SE blocks to strengthen channel-wise representation, with the goal of enhancing the accuracy and reliability of disease classification in medical imaging: leftmargin=1.5cm

- **Stage 1: Early Stage – Local Feature Extraction**
  Input: $X \in \mathbb{R}^{H\times W\times 3}$, the raw RGB image tensor.
  Operation: Depthwise convolution is applied independently on each input channel:

$$\mathbb{F}_1 = \text{DWConv}(X) = X * K_d$$

  where $*$ denotes convolution and K_d is a channel-specific depthwise kernel.
  This operation preserves spatial dimensions while allowing the network to learn fine-grained spatial details such as edges, gradients, and textural patterns.
  Output: $\mathbb{F}_1 \in \mathbb{R}^{H_1\times W_1\times C_1}$
  Purpose:
  - Extract low-level features from the input image.
  - Provide rich local information crucial for detecting early-stage or fine-grain pathological changes in kidney tissues.
- **Stage 2: Intermediate Stage – Contextual Feature Aggregation**
  Operation: Applies Layer Normalization to stabilize training and aggregate spatial information:

$$\mathbb{F}_2 = \text{LayerNorm}(\mathbb{F}_1) = \gamma \cdot \frac{\mathbb{F}_1 - \mu}{\sqrt{\sigma^2 + \varepsilon}} + \beta$$

  where:
  - μ and σ^2 are the mean and variance of each feature map channel.
  - γ and β are learnable scale and shift parameters.

  Output: $\mathbb{F}_2 \in \mathbb{R}^{H_2\times W_2\times C_2}$
  Purpose:
  - Aggregate mid-level contextual features.
  - Encode patterns such as tissue boundaries, shapes, and broader spatial relationships.
- **Stage 3: Final Stage – Semantic Feature Abstraction**

[1]Reused from IEEE Transactions on Pattern Analysis and Machine Intelligence, Vol 42, Issue 8, with permission from IEEE (License Number 6087661137132).

Operation: Applies LayerNorm to high-level features captured by deep convolutional blocks. This provides semantic abstraction of the full kidney structure.
Output: $\mathbb{F}_3 \in \mathbb{R}^{H_3 \times W_3 \times C_3}$
Purpose:
- Extract abstract and global representations.
- Distill complex anatomical and pathological features into compact, discriminative embeddings useful for classification.

- **SE Block**: The SE block enhances the representational capacity of the network by modeling interdependencies between feature channels. For each stage output $\mathbb{F}_i$, $i \in \{1, 2, 3\}$:
1. Squeeze: Spatial information is aggregated via global average pooling (GAP):

$$z_i = \frac{1}{H_i W_i} \sum_{h=1}^{H_i} \sum_{w=1}^{W_i} \mathbb{F}_i(h, w, :) \quad \text{where } z_i \in \mathbb{R}^{C_i}$$

2. Excitation: Adaptive reweighting is performed through two fully connected layers with non-linear activations:

$$s_i = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot z_i))$$

where:
- $W_1 \in \mathbb{R}^{\frac{C_i}{r} \times C_i}$, $W_2 \in \mathbb{R}^{C_i \times \frac{C_i}{r}}$
- σ: Sigmoid activation function

3. Recalibration: Feature maps are scaled channel-wise by the learned weights:

$$\tilde{\mathbb{F}}_i = s_i \odot \mathbb{F}_i$$

Purpose:
- Highlight diagnostically significant channels.
- Suppress irrelevant or noisy features.
- Improve the focus of the network on regions indicative of kidney pathology.

- **Multi-Scale Feature Fusion and Classification**:
Step 1: Global Average Pooling
Each recalibrated feature map is pooled to form a fixed-length vector:

$$f_i = \text{GAP}(\tilde{\mathbb{F}}_i), \quad f_i \in \mathbb{R}^{C_i}$$

Step 2: Feature Fusion
Multi-scale features are concatenated:

$$F = \text{Concat}(f_1, f_2, f_3) \in \mathbb{R}^{C_1 + C_2 + C_3}$$

Step 3: Fully Connected Classifier
The fused vector F passes through a dense classification head:

$$h_1 = \text{ReLU}(W_h F + b_h)$$
$$h_2 = \text{Dropout}(h_1)$$
$$y = \text{Softmax}(W_y h_2 + b_y)$$

Output: $y \in \mathbb{R}^4$ – a probability vector over the 4 kidney disease categories.
Purpose:
- Integrate information from multiple abstraction levels.
- Enhance classification robustness and accuracy.

4) FIVE ADDITIONAL MODELS

In this research, five additional models are used to compare the experimental results. These are: VGG19 [31], EfficientNet-B0 [32], Xception [33], MobileNet [34], DenseNet121 [35]. The comparison of these models is shown in the following Table 3.

5) GRAD-CAM

Grad-CAM is a visualization method that emphasizes specific regions in an image crucial for a DL model's decision-making process. When applied to CKDs detection, Grad-CAM aids in the decision-making process of the proposed model. Grad-CAM produces heatmaps that highlight the areas of the input image most influential in the model's predictions. This visualization aids in understanding which features the network emphasizes for each class, thereby enhancing the analysis of the model's behavior. This capability is particularly valuable in healthcare, as comprehending the rationale behind a model's prediction is essential for building trust and validation among medical professionals. Table 4 shows the outcomes of the Grad-CAM appraoch over CTK dataset. The steps of Grad-CAM algorithm are as follows [36]:

1) **Forward Pass to Obtain Class Scores**: The input image is processed through the CNN model using a standard forward pass to compute the class scores for each potential output. For instance, if the objective is to identify the presence of CKDs, the model will generate a score reflecting the probability of CKDs being present.
2) **Identify the Target Convolutional Layer**: Grad-CAM typically focuses on the final convolutional layer of a model, as this layer preserves the most spatial and semantic information regarding image features. The purpose is to analyze the activation patterns within this layer, which are crucial for recognizing class-specific details.
3) **Compute Gradients of the Target Class**: Gradients are calculated in relation to the target class (Cyst, Normal, Stone, or Tumor). These gradients illustrate the influence of each pixel in the feature maps on the target score, thereby highlighting the significance of each feature map for the specific class.
4) **Weighting Feature Maps with Gradients**: Gradients are averaged for each feature map to assign a weight to each, reflecting the significance of individual feature maps in predicting the target class.

TABLE 3. Comparison of the other five models.

Model	Key Features	Design Highlights	Advantages
VGG19	Deep CNN with uniform architecture using 3×3 convolutions	16 Conv + 3 FC layers; 5 max pooling layers; ReLU activations	Simplicity, high accuracy, widely adopted
EfficientNet-B0	Compound scaling of depth, width, and resolution	Depthwise separable convs; SE blocks; Swish activation	High accuracy with fewer parameters
Xception	Extreme Inception using depthwise separable convolutions	Entry, middle, and exit flows; residual connections; no FC layers	Efficient and accurate; fully convolutional
MobileNet	Lightweight CNN for mobile/embedded systems	Depthwise separable convs; width & resolution multipliers	Low latency and resource-efficient
DenseNet121	Densely connected layers enabling feature reuse	Each layer receives input from all previous layers; dense blocks with bottlenecks	Fewer parameters; mitigates vanishing gradients

5) **Creating the Heatmap**: A weighted combination of feature maps is calculated using the importance weights derived from the gradients. This highlights the most critical regions within the feature maps for predicting the target class, resulting in an activation map that identifies areas of the input image with the greatest influence on the model's prediction.
6) **Overlaying the Heatmap on the Input Scan**: The created heatmap is adjusted to fit the dimensions of the input image and placed on top of the original image, effectively emphasizing the areas most significant to the model's decision-making process.

C. EXPLAINABILITY AND MULTI-LAYER VISUALIZATION ANALYSIS

Figure 8 shows how feature representation becomes more refined across the network's hierarchy. In the early convolutional blocks (Stage 0), activation maps highlight low-level textural and edge details, such as tissue boundaries and variations in renal parenchyma intensity, which are crucial for understanding structural context. In the intermediate layers (Stage 2), activations focus more on morphologically relevant areas, reflecting the model's sensitivity to texture irregularities and subtle contrast differences often linked to tumor infiltration. In the deep semantic layers (Stage 3), the model's attention targets the most discriminative regions corresponding to tumor masses or abnormal tissue growths, demonstrating semantic abstraction of pathological cues.

The combined Grad-CAM map integrates information from all stages, visually confirming that the model uses multi-scale spatial cues—capturing both local textural irregularities and global contextual semantics crucial for accurate kidney cancer classification. This visualization provides concrete evidence supporting the model's design rationale, beyond textual description or theoretical claims.

IV. RESULTS OVERVIEW AND ANALYSIS

A. MEASUREMENT OF EFFECTIVENESS

The performance of the six DL models- SAFF-ConvNeXt, DenseNet121, EfficientNet-B0, MobileNet, VGG19, and

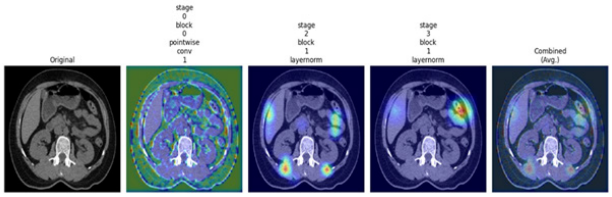


FIGURE 8. The Multi-layer Grad-CAM visualization of the SAFF-ConvNeXt model for CKD detection.

Xception - is assessed using Equations 10 through 15 [37]:

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + FP + TN + FN)} \tag{10}$$

$$\text{Precision} = \frac{TP}{(TP + FP)} \tag{11}$$

$$\text{Sensitivity} = \frac{TP}{(TP + FN)} \tag{12}$$

$$\text{Specificity} = \frac{TN}{(TN + FP)} \tag{13}$$

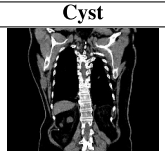
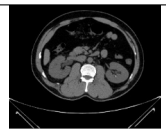
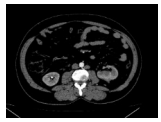
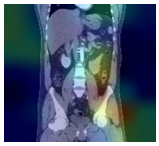
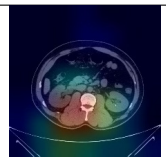
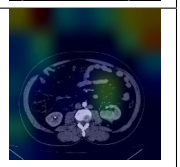
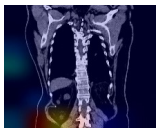
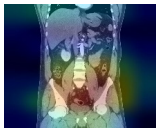
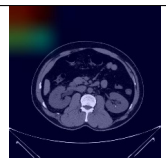
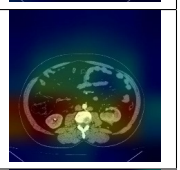
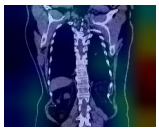
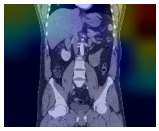
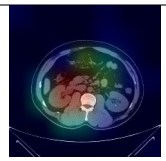
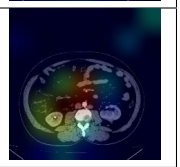
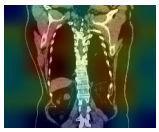
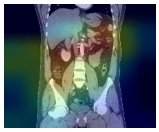
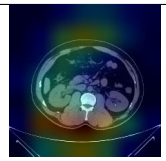
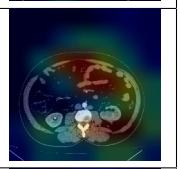
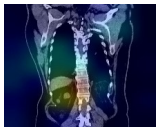
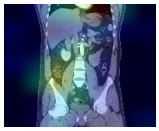
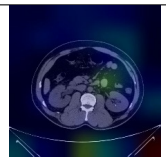
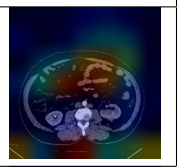
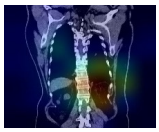
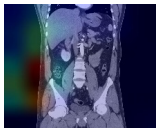
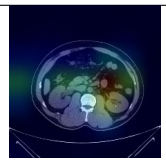
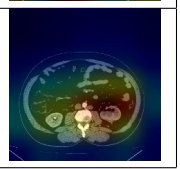
$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{14}$$

$$\text{False Negative Rate (FNR)} = \frac{FN}{TP + FN} \tag{15}$$

$$\text{Negative Predictive Value (NPV)} = \frac{TN}{TN + FN} \tag{16}$$

True Positives (TP) are correctly predicted positive cases, while **False Positives (FP)** are instances wrongly classified as positive. **True Negatives (TN)** are accurate negative predictions, and **False Negatives (FN)** occur when positive instances are misclassified as negative. Accuracy measures a model's overall prediction correctness but may not perform well with imbalanced data. Precision assesses the reliability of positive predictions, with a range of 0 to 1. Sensitivity (or recall) evaluates the model's ability to correctly identify positive cases. The **F1-score** balances precision and recall, ranging from 0 to 1, offering a holistic performance view. **False Negative Rate (FNR)** and **Negative**

TABLE 4. The outputs of the Grad-CAM technique over CTK dataset.

Model/Class	Cyst	Normal	Stone	Tumor
Original image				
SAFF-ConvNeXt				
DenseNet121				
EfficientNet-B0				
MobileNet				
VGG19				
Xception				

Predictive Value (NPV) are specific to the model's handling of negative predictions: FNR assesses missed positive instances, while NPV reflects the accuracy of negative predictions.

B. COMPUTATIONAL EFFICIENCY

From Table 5, the proposed SAFF-ConvNeXt model demonstrated marked advantages over conventional architectures such as DenseNet121, EfficientNet-B0, MobileNet, VGG19, and Xception. Despite containing only 4.6×10^6 trainable parameters—a relatively small fraction of its total parameters—the model attains the highest performance across all evaluated baselines. This parameter efficiency facilitates effective transfer learning from pre-trained weights and targeted fine-tuning on the task dataset, reducing both the risk of overfitting and training time. The combination of multi-scale feature fusion and attention mechanisms enables the network to prioritize the most informative image regions and to capture subtle patterns that lighter architectures (e.g., MobileNet, EfficientNet-B0) and fully trainable models (e.g., VGG19, Xception) may overlook. Although the model's inference throughput (approximately 0.47 FPS) is lower than that of simpler alternatives, its superior discriminative power and robust feature representations make it particularly well-suited to high-accuracy applications, notably in complex medical imaging tasks. Overall, the architecture balances efficient fine-tuning, attention-guided feature refinement, and multi-scale fusion, establishing it as the most effective model among those assessed.

C. EXPERIMENTAL FRAMEWORK

In this study, we conducted an experiment on the CTK dataset for multi-classification using the Kaggle environment utilizing an NVIDIA Tesla T4 GPU with 16 GB of

TABLE 5. Comparison of the six DL models based on parameters, FPS, and Training time.

Model	Total Params	Trainable Params	Non-Trainable Params	FPS	Training Time
SAFF-ConvNeXt	75,318,660	4,596,996	70,721,664	0.47	8m 37.94s
DenseNet121	7,564,356	7,480,708	83,648	2.24	14m 0.70s
EfficientNetB0	4,707,495	4,665,472	42,023	3.46	7m 57.53s
MobileNet	3,755,716	3,733,828	21,888	11.97	5m 38.95s
VGG19	20,289,092	20,289,092	0	36.93	5m 39.12s
Xception	21,912,620	21,858,092	54,528	7.27	10m 40.66s

TABLE 6. The hyperparameter values used in our experiment.

Parameter	Value
img_size	224 x 224
Number of epochs	20
Batch size (BS)	32
Activation	softmax
Dropout	0.05
Optimizer	Adam
Initial learning rate (LR)	0.001
loss	categorical_crossentropy
Patience	10

VRAM. The environment was powered by a 13th Gen Intel Core i9-13900H CPU running at 2.60 GHz and equipped with 32 GB of RAM. We used Python programming language along with the TensorFlow library. TensorFlow is a widely adopted DL architecture developed by Google. TensorFlow is extensively employed to build and deploy DL and ML models, particularly in the areas of DL and AI. Furthermore, the hyperparameters utilized in our experiment, along with their respective values, were listed in Table 6.

D. PROPOSED SAFF-ConvNeXt MODEL EVALUATION

In our research, we divided the imbalanced CTK dataset into a training set and a testing set. The training set contained 85% of the total data, amounting to 10,579 samples, while the testing set represented the remaining 15%, which included 1,867 instances. We utilized the undersampling technique to reduce the training set from 10,579 to 2,000 CT scans. To harness the advantages of transfer learning, we pre-trained six DL models. The initial phase involved supervised pre-training, where these models were trained using the ImageNet dataset. Subsequently, training was conducted on the CTK dataset. Lastly, we assessed the performance metrics of the six DL approaches using a series of equations (10 - 16) at the conclusion of the experiment.

We conducted a multi-class classification study on the CTK dataset using six DL models: SAFF-ConvNeXt, DenseNet121, EfficientNet-B0, MobileNet, VGG19, and Xception. The dataset contains four classes: Cyst, Normal, Stone, and Tumor. The performance of five iterations (folds) of each model, as well as the average of the evaluation metrics for the six DL models, are presented in Tables 7 through 12. In each iteration, one of the five folds was used as the validation set, and the other four folds were used as the training set. Each model was trained on the training set and evaluated or tested on the validation set. This process was repeated five times so that each fold was used as the validation set exactly once. After the five iterations, we calculated the average of the performance metrics over the five folds to get an overall estimate of model performance.

The results of the models' average accuracy across all classes evaluation were as follows: the SAFF-ConvNeXt achieved an accuracy rate of 99.75%, the DenseNet121 model achieved an accuracy rate of 97.48%, the EfficientNet-B0 model achieved an accuracy rate of 98.78%, the MobileNet model achieved an accuracy rate of 97.55%, the VGG19 model achieved an accuracy rate of 98.55%, and the Xception model achieved an accuracy rate of 97.79%. Based on these findings, the SAFF-ConvNeXt demonstrated the highest level of accuracy among the evaluated models.

DenseNet121 had 98.28% specificity, 6.56% FNR, 98.32% NPV, 94.10% precision, 93.44% recall, and 93.46% F1-score averages. EfficientNet-B0 recorded 99.26% specificity, 2.35% FNR, 99.13% NPV, 96.01% precision, 97.65% recall, and 96.71% F1-score averages. MobileNet achieved 98.38% specificity, 5.55% FNR, 98.35% NPV, 93.88% precision, 94.45% recall, and 93.93% F1-score averages. VGG19 achieved 99.04% specificity, 3.04% FNR, 99.01% NPV, 96.18% precision, 96.96% recall, and 96.43% F1-score averages. Xception achieved 98.50% specificity, 5.43% FNR, 98.47% NPV, 94.81% precision, 94.57% recall, and 94.60% F1-score averages.

The proposed approach demonstrated exceptional performance, attaining high scores across various evaluation metrics. Its specificity, precision, F1-score, NPV, recall, and accuracy all surpassed 99%, highlighting its remarkable abilities. Additionally, the model showed an impressively low (FNR) of only 0.61%. These exceptional performance results across a range of assessment criteria strongly position the SAFF-ConvNeXt as a highly effective and dependable tool for multi-classification tasks.

Figures 9, 10, 11, 12, 13, and 14 show the performance of the SAFF-ConvNeXt, DenseNet121, EfficientNet-B0, MobileNet, VGG19, and Xception models during training and validation across 5-fold cross-validation, respectively. The figures are divided into four subplots: training accuracy, training loss, validation accuracy, and validation loss, for each fold (fold_1 to fold_5).

From Figure 9, for the **SAFF-ConvNeXt** model, the training accuracy rises sharply across all folds in the initial epochs, approaching nearly 100%, while the training loss declines rapidly to almost zero. This indicates quick and efficient learning with minimal underfitting. The validation accuracy also demonstrates a strong upward trend across

TABLE 7. The average results of the SAFF-ConvNeXt after the five iterations on the CTK dataset.

	Fold	Acc (%)	Spec (%)	FNR (%)	NPV (%)	Prec (%)	Rec (%)	F1 (%)
SAFF-ConvNeXt	0	99.813	99.866	0.710	99.886	99.671	99.290	99.477
	1	99.920	99.939	0.315	99.954	99.878	99.685	99.785
	2	99.866	99.910	0.229	99.900	99.577	99.771	99.673
	3	99.759	99.837	0.717	99.848	99.381	99.284	99.332
	4	99.411	99.622	1.114	99.544	98.601	98.886	98.725
	Avg	**99.754**	**99.835**	**0.617**	**99.827**	**99.422**	**99.383**	**99.397**

TABLE 8. The average results of the DenseNet121 after the five iteration on the CTK dataset.

	Fold	Acc (%)	Spec (%)	FNR (%)	NPV (%)	Prec (%)	Rec (%)	F1 (%)
DenseNet121	0	97.70	98.34	7.61	98.52	95.25	92.39	93.50
	1	97.62	98.52	4.21	98.23	92.96	95.79	94.12
	2	96.30	97.27	11.83	97.81	94.16	88.17	90.39
	3	98.07	98.68	4.70	98.68	95.13	95.30	95.21
	4	97.72	98.60	4.43	98.35	93.02	95.57	94.09
	Avg	**97.48**	**98.28**	**6.56**	**98.32**	**94.10**	**93.44**	**93.46**

TABLE 9. The average results of the EfficientNet-B0 after the five iterations on the CTK dataset.

	Fold	Acc (%)	Spec (%)	FNR (%)	NPV (%)	Prec (%)	Rec (%)	F1 (%)
EfficientNet-B0	0	99.04	99.42	1.79	99.31	96.60	98.21	97.30
	1	99.38	99.64	1.13	99.56	97.72	98.87	98.26
	2	98.85	99.32	2.12	99.13	96.02	97.88	96.86
	3	98.88	99.25	2.49	99.24	96.78	97.51	97.09
	4	97.78	98.66	4.22	98.41	92.94	95.78	94.01
	Avg	**98.78**	**99.26**	**2.35**	**99.13**	**96.01**	**97.65**	**96.71**

TABLE 10. The average results of the MobileNet after the five iterations on the CTK dataset.

	Fold	Acc (%)	Spec (%)	FNR (%)	NPV (%)	Prec (%)	Rec (%)	F1 (%)
MobileNet	0	97.59	98.30	6.64	98.45	94.60	93.36	93.85
	1	97.88	98.63	4.89	98.60	93.97	95.11	94.32
	2	98.71	99.10	3.16	99.15	96.96	96.84	96.89
	3	96.65	97.81	7.40	97.80	91.10	92.60	91.31
	4	96.92	98.07	5.67	97.73	92.79	94.33	93.26
	Avg	**97.55**	**98.38**	**5.55**	**98.35**	**93.88**	**94.45**	**93.93**

each fold, stabilizing between 98% and 100%, reflecting the model's excellent generalization capability. Furthermore, the validation loss consistently decreases throughout the training process, with all folds achieving values below 0.1. Although slight fluctuations are noted in some folds (particularly folds 2 and 3), these variations are typical in k-fold training and do not considerably impact overall performance. The uniform behavior across folds reinforces the reliability of the proposed architecture, its resistance to overfitting, and its ability to learn distinguishing features for accurate classification of CKDs.

From Figure 10, the **DenseNet121** model demonstrated a steady increase in training accuracy across all folds, starting at around 0.4 and reaching near or above 0.9, with mostly smooth accuracy curves indicating effective learning. While there was some minor variation among the folds,

TABLE 11. The average results of the VGG19 after the five iterations on the CTK dataset.

	Fold	Acc (%)	Spec (%)	FNR (%)	NPV (%)	Prec (%)	Rec (%)	F1 (%)
VGG19	0	99.09	99.40	2.13	99.36	97.35	97.87	97.60
	1	97.86	98.62	4.30	98.52	93.84	95.70	94.42
	2	99.28	99.48	2.01	99.53	98.33	97.99	98.16
	3	99.38	99.62	1.28	99.54	98.15	98.72	98.42
	4	97.13	98.05	5.49	98.10	93.23	94.51	93.54
	Avg	**98.55**	**99.04**	**3.04**	**99.01**	**96.18**	**96.96**	**96.43**

TABLE 12. The average results of the Xception after the five iterations on the CTK dataset.

	Fold	Acc (%)	Spec (%)	FNR (%)	NPV (%)	Prec (%)	Rec (%)	F1 (%)
Xception	0	97.35	98.26	5.90	98.12	93.03	94.10	93.52
	1	98.45	98.99	3.82	98.93	95.54	96.18	95.84
	2	98.15	98.77	4.20	98.74	95.43	95.80	95.56
	3	98.29	98.75	5.13	98.89	97.03	94.87	95.85
	4	96.71	97.75	8.12	97.69	93.00	91.88	92.24
	Avg	**97.79**	**98.50**	**5.43**	**98.47**	**94.81**	**94.57**	**94.60**

the model achieved high accuracy overall. Training loss dropped significantly from above 3.5 to below 0.5 by the 10th epoch and stabilized after 10-15 epochs, reflecting effective loss minimization. Consistency was good across folds, with all converging to similarly low loss values by the end of training. The validation accuracy initially began at a lower level compared to the training accuracy but improved steadily over the epochs, with noticeable variations across different folds, particularly in later epochs. While some folds achieved accuracy close to or above 0.95, others, like fold_3 and fold_4, showed inconsistent performance. This indicates that the model generalized well on certain folds but struggled with others. Similarly, although the validation loss decreased overall, it exhibited significant fluctuations, particularly in folds 2 and 4, which had higher losses at times. These variations suggest uneven model performance across the validation folds.

The model performed well during training, achieving high accuracy and low loss across all folds. However, validation performance varied, indicating potential overfitting to training folds and challenges with certain validation sets. The fluctuations in the validation metrics suggested inconsistent generalizability, likely due to differences in data difficulty across folds. Although the model converged during training, with metrics stabilizing after 10-15 epochs, validation metrics remained unstable, hinting at a need for further tuning, such as regularization or additional epochs. The significant gap between high training accuracy and variable validation accuracy raised concerns about overfitting, suggesting the model may struggle to generalize to unseen data.

From Figure 11, the **EfficientNet-B0** model demonstrated a steady increase in training accuracy across all folds, starting at approximately 0.6 and approaching 1.0 by the 15th epoch, with smooth and consistent curves showing minimal variation. The training loss rapidly decreased from around 1.2 to near zero within the first few epochs, stabilizing after about 5 epochs. By the end of training, all folds converged to very low loss values, indicating effective loss minimization and uniform learning from the training data across different subsets. The validation accuracy started at approximately 0.75 and improved steadily, with most folds exceeding 0.95 by the end of training. However, there were variations among the folds; for instance, fold_1 achieved near-perfect accuracy, while fold_3 showed more fluctuation. This overall upward trend indicated good generalization, though some instability was noted compared to the training set. Validation loss decreased quickly, mirroring training loss, and stabilized after about five epochs, but it fluctuated more, particularly in folds like fold_5. Despite these variations, the overall reduction in validation loss suggested an improving ability to generalize across the data.

The EfficientNet-B0 model exhibited strong training performance, achieving near-perfect accuracy and low training loss. While validation performance was also robust, it showed variability, particularly in later epochs, indicating sensitivity to specific validation data subsets. The model stabilized after about five epochs, with a generally positive trend, though some folds experienced fluctuations. Although training accuracy approached 100%, validation accuracy did not consistently reach that level, and validation loss was more

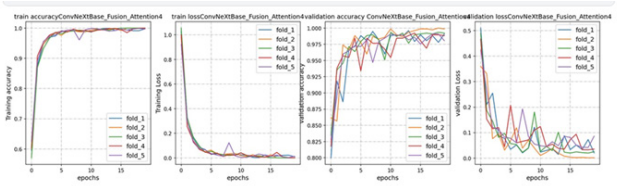


FIGURE 9. The accuracy and loss of the five folds cross validation for the SAFF-ConvNeXt model.

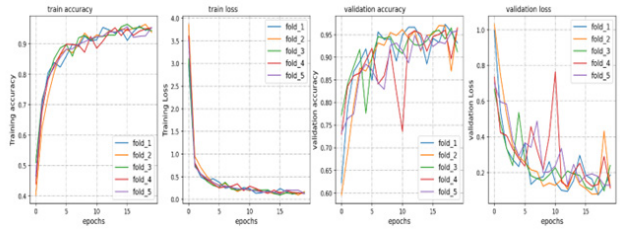


FIGURE 10. The accuracy and loss of the five folds cross validation for the DenseNet121 model.

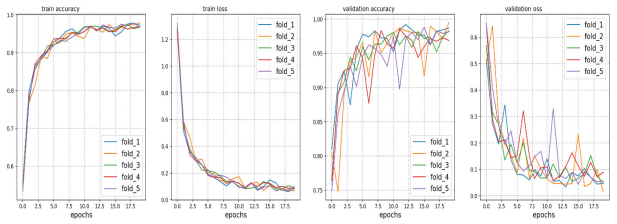


FIGURE 11. The accuracy and loss of the five folds cross validation for the EfficientNet-B0 model.

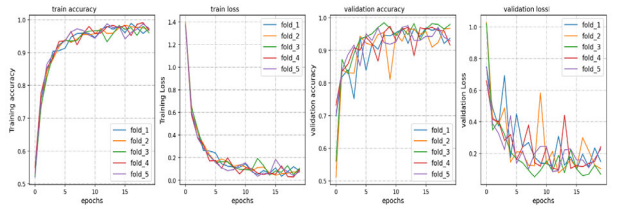


FIGURE 12. The accuracy and loss of the five folds cross validation for the MobileNet model.

variable, suggesting potential overfitting to the training data, despite the model's strong overall performance.

As shown in Figure 12, the **MobileNet** model showed rapid increases in training accuracy, stabilizing between 0.9 and 1.0 by epoch 10 across all five folds, with minimal variation, indicating consistent performance. Training loss sharply decreased to around 0.1 to 0.2 after 10 epochs, with close convergence among folds, reflecting efficient learning. Validation accuracy also rose quickly, stabilizing around 0.85 to 0.95 after 10 to 15 epochs, though with more variability across folds; fold_2 and fold_5 performed slightly better later on. Validation loss declined similarly but exhibited more fluctuations, with spikes in certain folds

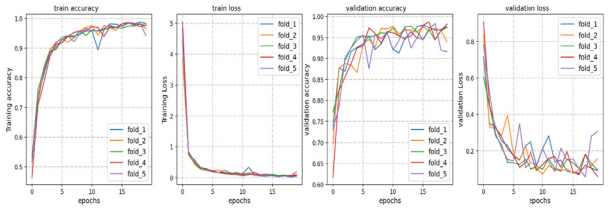


FIGURE 13. The accuracy and loss of the five folds cross validation for the VGG19 model.

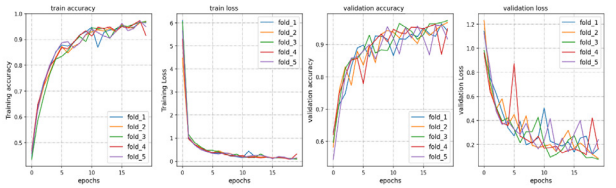


FIGURE 14. The accuracy and loss of the five folds cross validation for the Xception model.

suggesting potential overfitting. By the end of training, validation loss settled between 0.2 and 0.4 for most folds. Overall, the model demonstrated effective learning, with steady improvements in accuracy and decreasing losses, though some folds faced more challenging validation data. The training process exhibited characteristics of successful generalization, with consistent performance across folds.

As shown in Figure 13, the **VGG19** model exhibited strong performance during training, with training accuracy rapidly rising and stabilizing near 1.0 by epoch 10 across all five folds, indicating consistent learning. The training loss started high but decreased sharply to around 0.1, showing effective learning. Validation accuracy began at approximately 0.6 and increased to between 0.9 and 0.95 by epoch 10, with slight variations among folds; folds 2 and 5 reached close to 0.95, while fold 3 was slightly lower at around 0.90. Validation loss decreased from 0.8 to 1.0, stabilizing between 0.1 and 0.3, with some folds (notably fold 3 and fold 5) showing higher losses, suggesting potential overfitting. Overall, VGG19 demonstrated rapid learning, robust generalization across folds, and minimal variations, outperforming MobileNet in stability despite slightly higher fluctuations in validation loss.

The **Xception** model, as shown in Figure 14, showed a steady increase in training accuracy, reaching close to 1.0 by epochs 10-15, with minimal variation across folds, indicating effective learning. Training loss started high but quickly dropped to near 0 by epoch 10, with similar trends across folds. Validation accuracy began around 0.6, rapidly rising to between 0.9 and 1.0, although some variability was noted after epoch 10, particularly in folds like fold_2 and fold_4. Validation loss decreased from 1.2 within the first few epochs, though it exhibited more variability in later stages, with folds such as fold_3 and fold_5 showing larger spikes. Overall, the

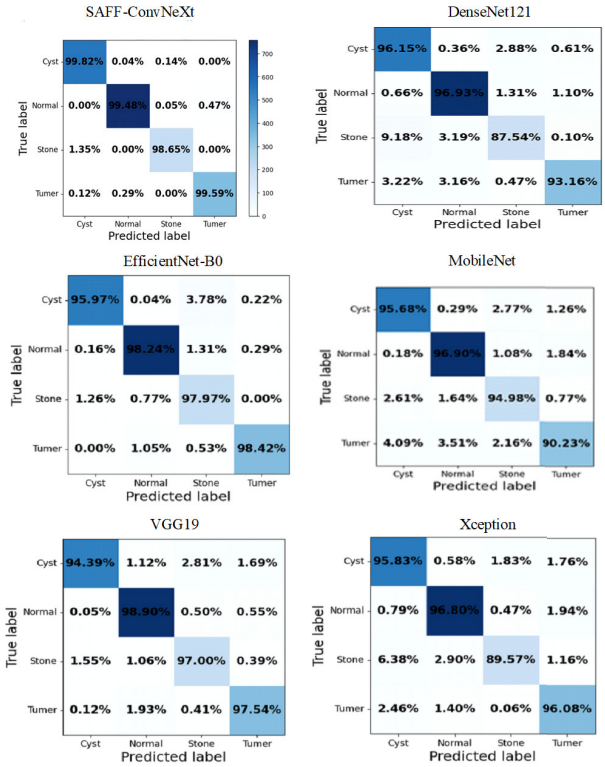


FIGURE 15. The six DL confusion matrices for CKDs classification.

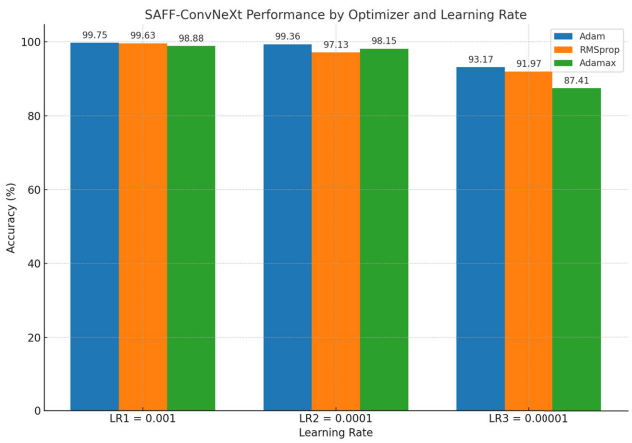


FIGURE 16. SAFF-ConvNeXt Model's accuracies with three optimizers and LR values.

model demonstrated rapid and effective learning, with strong generalization despite some validation variability. The close alignment of training metrics suggested reliable performance across different data subsets, although further tuning might be needed to enhance generalization. In summary, the Xception model achieved high accuracy and low loss, reflecting robust learning capabilities.

The confusion matrices in Figure 15 present the effectiveness of the six DL approaches — SAFF-ConvNeXt, DenseNet121, EfficientNet-B0, MobileNet, VGG19, and Xception—after five-fold cross-validation on the CTK

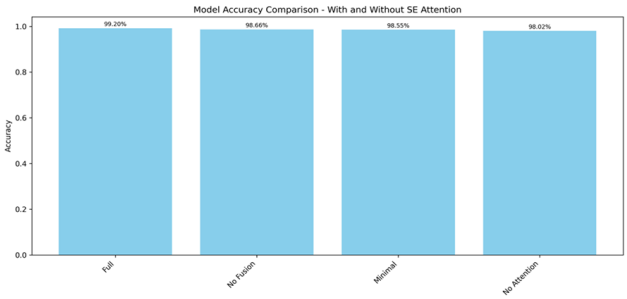


FIGURE 17. SAFF-ConvNeXt's accuracy comparison- with and without Attention & Fusion.

dataset, classifying four classes: **Normal, Cyst, Tumor and Stone**. SAFF-ConvNeXt achieves exceptional accuracy, leading in all categories: **Cyst** (99.82%), **Normal** (99.48%), **Stone** (98.65%), and **Tumor** (99.59%), with very few misclassifications ($\leq$ 1.35%). EfficientNet-B0 and VGG19 also perform well, though slightly behind, especially in the Stone category (EfficientNet: 97.97%; VGG19: 97.00%) and Tumor (EfficientNet: 98.42%; VGG19: 97.54%).

In contrast, DenseNet121, MobileNet, and Xception face considerable challenges with **Stone** (87.54%, 94.98%, 89.57%) and **Tumor** (93.16%, 90.23%, 96.08%), frequently misclassifying these as **Cyst** or **Normal**. For instance, DenseNet121 misclassifies Stone as Cyst 9.18% of the time, while MobileNet confuses Tumor with Cyst/Normal at a combined rate of 7.6%.

In summary, **SAFF-ConvNeXt** stands out as the best performer, surpassing the others by more than 1.5% in each category, while lighter models (MobileNet, Xception) and DenseNet121 show significant deficiencies in recognizing Stone and Tumor.

E. ABLATION STUDY

An ablation study is an essential technique utilized in research, especially in ML and DL. This method aims to assess the significance of particular elements or factors within a model, algorithm, or system. By methodically removing or modifying these elements and analyzing how performance is affected, we can gain insight into the function and importance of each component. This process provides empirical evidence to explain why certain design decisions work effectively. Ablation studies are particularly beneficial for refining model architectures, clarifying the behavior of complex systems, and ensuring reproducibility and transparency in research. Overall, ablation studies are vital for improving understanding and performance in ML and DL applications [38].

In this research, we performed an ablation analysis through two experiments. In the first experiment, we adjusted the optimizers and LR settings. Initially, we used the Adaptive Moment Estimation (Adam) optimizer with a LR of 0.001, achieving an accuracy of 99.754%. We then repeated this with two additional optimization methods: Root Mean Square

TABLE 13. The SAFF-ConvNeXt accuracies with three optimizers and LR values.

	LR Value	Optimizer		
		Adam	RMSprop	Adamax
SAFF-ConvNeXt	LR1 = 0.001	**99.754**	99.625	98.875
	LR2 = 0.0001	**99.357**	97.134	98.152
	LR3 = 0.00001	**93.170**	91.965	87.413

TABLE 14. SAFF-ConvNeXt's accuracy comparison- with and without Attention & Fusion.

Variant	Attention	Fusion	Accuracy	Loss
Full Model	✓	✓	**99.754**	**2.095**
No Fusion	✓	✗	98.661	6.141
Minimal	✗	✗	98.554	5.347
No Attention	✗	✓	98.018	6.061

Propagation (RMSprop) and AdamX, a variant of the Adam optimizer. For all three optimizers, we evaluated the LR of 0.001, 0.0001, and 0.00001. The results, which show the accuracy of the SAFF-ConvNeXt model across the different optimizers and learning rates, are presented in Table 14 and illustrated in Figure 16. These findings indicate that the Adam optimizer with a learning rate of 0.001 achieved the highest accuracy in our assessment.

In the second step, we examined the effect of SE blocks as attention modules and a multi-scale feature fusion approach. Table 14 and Figure 17 present the results of the second experiment. Table 14 and Figure 17 show that The Full Model, which incorporates both components, achieved the highest accuracy of 99.754% and the lowest loss of 2.095 %, indicating that the combination of these design elements provides complementary benefits to feature representation and learning stability. Removing the fusion component while retaining attention (No Fusion) resulted in a noticeable drop in accuracy to 98.661 % and a sharp increase in loss to 6.141 %, suggesting that fusion plays a key role in integrating multi-scale or multi-level features effectively. Similarly, the Minimal configuration, which removes both attention and fusion, performed slightly worse than No Fusion in accuracy (98.554%) but achieved a marginally better loss (5.347%), indicating that removing attention alone can sometimes reduce overfitting but at the cost of peak accuracy.

When attention was removed but fusion was retained (No Attention), the model's accuracy further decreased to 98.018% with a loss of 6.061%, reinforcing the value of attention in refining feature focus and improving classification precision.

Overall, the results confirm that both attention and fusion contribute significantly to performance, with their combined use producing the most accurate and stable model configuration.

F. STATISTICAL ASSESSMENT OF THE RESULTS

To assess the reliability of the SAFF-ConvNeXt's outcomes, a thorough statistical analysis was carried out. This analysis examined the variance, standard deviation, and standard error of the mean (SEM) of the results. The observed measurements were gathered following 5-fold cross validation of the process. The comprehensive examination of the SAFF-ConvNeXt for the CTK dataset is provided in Tables 15 to 18. From Table 15, SAFF-ConvNeXt achieved the highest average performance at 99.75%, indicating it was highly accurate and consistent. EfficientNet-B0 followed with an average of 98.78%. SAFF-ConvNeXt had the lowest variance (0.032251) and standard deviation (0.179586), suggesting it was very stable with minimal fluctuation across the folds. VGG19 had the highest variance (0.79967) and standard deviation (0.894243), implying more variability and inconsistent performance. SAFF-ConvNeXt had the smallest standard error of the mean (0.0803133), indicating its mean estimate was very reliable. VGG19 had the largest standard error of the mean (0.399917), suggesting more uncertainty around its average performance. SAFF-ConvNeXt had the narrowest range between its maximum (99.92%) and minimum (99.41%) scores, showing high reliability across all folds. MobileNet had a relatively larger range (Maximum: 98.71 %, Minimum: 96.65%) and higher variability, reflected in its high standard deviation. In summary, SAFF-ConvNeXt was the top-performing model with the highest accuracy, lowest variability, and most reliable performance estimate. EfficientNet-B0 and VGG19 also showed strong performance, but VGG19 had higher variability, making its performance less consistent. MobileNet and Xception showed relatively good results but were less stable compared to the SAFF-ConvNeXt model.

From Table 16, SAFF-ConvNeXt demonstrated the highest average performance at 99.40%, showcasing minimal fluctuations. EfficientNet-B0 followed closely with a strong average of 96.71%, while VGG19 also exhibits a high average of 96.43%. Notably, SAFF-ConvNeXt exhibited the lowest variance (0.137070) and standard deviation (0.370229), indicating exceptional consistency and minimal variability in its performance. In contrast, VGG19 had the highest variance (4.140874) and standard deviation (2.034914), suggesting a wider spread and greater fluctuations across the folds. SEM further highlighted SAFF-ConvNeXt's reliability, with the smallest SEM (0.165571), suggesting a highly precise and reliable estimate of its average performance. VGG19, on the

other hand, had the largest SEM (0.910041), indicating a less precise estimate due to its higher variability. The performance range also reflected SAFF-ConvNeXt's consistency, with the narrowest gap between its maximum (99.78%) and minimum (99.73%) scores across the folds. VGG19 and MobileNet, in comparison, displaied wider performance ranges, aligning with their higher variability. In summary, SAFF-ConvNeXt emerged as the best-performing and most consistent model, with high average accuracy and low variability. EfficientNet-B0 and VGG19 also performed well on average, but VGG19's high variability and wide performance range make its results less stable. Other models, such as DenseNet121, MobileNet, and Xception, exhibited relatively lower performance and higher variability compared to the SAFF-ConvNeXt model.

In Table 17, the confidence intervals (CIs) represent the range within which the true population mean is expected to fall with a high level of confidence (typically 95%). From Table 15, SAFF-ConvNeXt had the narrowest CI of [99.59621, 99.91103], reflecting a high degree of certainty and consistency in its performance. In contrast, VGG19 had a wider CI of [97.76464,99.33231], suggesting more variability and less certainty about its true performance. The margin of error indicates the range around the sample mean within which the true population mean lies. SAFF-ConvNeXt had the smallest margin of error of 0.157414, further confirming its reliable and consistent performance. VGG19, on the other hand, had the largest margin of error of 0.783838, showing that its performance is more variable and that there is less confidence in its true average accuracy. In summary, SAFF-ConvNeXt was the most reliable model in terms of performance consistency, precision (small standard error of the mean), and predictability (narrow CI and small margin of error). Its performance was both stable and easy to trust. VGG19, on the other hand, had the highest variability, leading to greater uncertainty in its true performance. Its wide confidence interval and large margin of error make it less predictable. MobileNet, EfficientNet-B0, and Xception lie in the middle, with more variability than SAFF-ConvNeXt but less than VGG19, showing moderate consistency but still subject to fluctuations in real-world applications.

From the data in Table 18, the SAFF-ConvNeXt model had the narrowest CI of [99.07289,99.72194], suggesting its true performance was highly predictable and consistent. In contrast, DenseNet121 had a much wider CI of [92.03194,94.8892], indicating more uncertainty in its performance. Similarly, VGG19 had a wide CI of [94.64625,98.21361], reflecting high variability and less certainty in its expected performance. The margin of error indicates how far the estimated mean could be from the true mean. SAFF-ConvNeXt had the smallest margin of error at 0.324521, meaning its performance estimate was highly accurate and predictable. On the other hand, VGG19 had the largest margin of error at 1.783681, reinforcing that its performance fluctuates and is less reliable compared to the other models. Overall, SAFF-ConvNeXt stood out with minimal variability, high precision, and narrow confidence intervals. Its margin of error was also the smallest, making it the most reliable and consistent model in the group. DenseNet121, MobileNet, and Xception showed moderate variability, with wider confidence intervals and larger margins of error compared to the SAFF-ConvNeXt. VGG19 had the highest variability, the widest confidence interval, and the largest margin of error, indicating that its performance is less consistent and more uncertain. In summary, the SAFF-ConvNeXt remains the most stable and dependable model, while VGG19 is the least consistent and has the most variability across folds.

After assessing the reliability of the SAFF-ConvNeXt's outcomes, a thorough statistical analysis examined the variance, standard deviation, and SEM of the results. Moreover, we implemented the Friedman Ranking Test with the Nemenyi post-hoc test, as shown below in Tables 19 and 20.

The Friedman test was calculated using the following formula:

$$\chi_F^2 = \frac{12}{nk(k+1)}\left(\sum_{j=1}^{k} R_j^2 - \frac{k(k+1)^2}{4}\right) \tag{17}$$

where $n = 1$, since there is only one group of measurements and $k = 6$ (number of models). Hence $\chi_F^2 = 5$. The critical difference can be calculated using the following formula:

$$CD = q_\alpha \sqrt{\frac{k(k+1)}{6n}} \tag{18}$$

For $k = 6$ at $\alpha = 0.05$, $q_{0.05}$ is approximately 2.5. Hence, $CD = 6.614$.

Next, we will compare $\chi_F^2 = 5$ with the $CD = 6.614$. Since $5 < 6.614$, we fail to reject the null hypothesis, indicating that there are no significant differences in model performance based on the accuracy values provided.

The Friedman Ranking Test did not find a significant difference at $\alpha = 0.05$ and frame the improvement as practically meaningful rather than strictly statistically substantial.

G. EXTERNAL VALIDATION

To demonstrate the robustness and generalization capability of the proposed model across varying data distributions and ensure its impact contribution in medical imaging, we tested the model on the Axial CT Imaging dataset for AI-Powered Kidney Stone [39]. This dataset contains CT scan images for kidney stone detection, organized into two groups: individuals diagnosed with kidney stones and individuals without the condition. The images were carefully selected, reviewed, and labeled by experienced medical professionals to ensure high quality and reliability for both research and educational use. The dataset was collected from medical centers in Sulaimani and Rania in the Kurdistan Region of Iraq, the dataset offers region-specific imaging information on kidney stone disease. It includes 3,364 original CT images and 35,457 augmented images, making it a valuable resource

TABLE 15. The statistical analysis of the six DL models based on the accuracy using the CTK dataset.

Models	Avg (%)	max (%)	Min (%)	σ^2	σ	SEM
SAFF-ConvNeXt	99.75	99.92	99.41	0.032251	0.179586	0.080313
DenseNet121	97.48	98.07	96.30	0.371519	0.609524	0.272587
EfficientNet-B0	98.78	99.38	97.78	0.289928	0.53845	0.240802
MobileNet	97.55	98.71	96.65	0.534356	0.730997	0.326912
VGG19	98.55	99.38	97.13	0.79967	0.894243	0.399917
Xception	97.79	98.45	96.71	0.435667	0.660051	0.295184

TABLE 16. The statistical analysis of the six DL models based on the F1-score using the CTK dataset.

Models	Avg (%)	max (%)	Min (%)	σ^2	σ	SEM
SAFF-ConvNeXt	99.40	99.78	98.73	0.137070	0.370229	0.165571
DenseNet121	93.46	95.21	90.39	2.656418	1.629852	0.728892
EfficientNet-B0	96.71	98.26	94.01	2.040775	1.428557	0.63887
MobileNet	93.93	96.89	91.31	3.251517	1.803196	0.806414
VGG19	96.43	98.42	93.54	4.140874	2.034914	0.910041
Xception	94.60	95.85	92.24	2.153494	1.467479	0.656276

TABLE 17. The confidence interval of the six DL models based on the accuracy using the CTK dataset.

Models	σ	SEM	Confidence interval	Margin of Error
SAFF-ConvNeXt	0.179586	0.080313	[99.59621, 99.91103]	0.157414
DenseNet121	0.609524	0.272587	[96.94832, 98.01686]	0.534271
EfficientNet-B0	0.53845	0.240802	[98.31217, 99.25612]	0.471972
MobileNet	0.730997	0.326912	[96.91148, 98.19297]	0.640747
VGG19	0.894243	0.399917	[97.76464, 99.33231]	0.783838
Xception	0.660051	0.295184	[97.20933, 98.36646]	0.57856

TABLE 18. The confidence interval of the six DL models based on the F1-score using the CTK dataset.

Models	σ	SEM	Confidence interval	Margin of Error
SAFF-ConvNeXt	0.370229	0.165571	[99.07289, 99.72194]	0.324521
DenseNet121	1.629852	0.728892	[92.03194, 94.8892]	1.428628
EfficientNet-B0	1.428557	0.63887	[95.45321, 97.95758]	1.252185
MobileNet	1.803196	0.806414	[92.34547, 95.50661]	1.580571
VGG19	2.034914	0.910041	[94.64625, 98.21361]	1.783681
Xception	1.467479	0.656276	[93.31545, 95.88805]	1.286302

TABLE 19. The average accuracies of the six DL models using the CTK dataset.

Models	Avg (%)
SAFF-ConvNeXt	99.75
DenseNet121	97.48
EfficientNet-B0	98.78
MobileNet	97.55
VGG19	98.55
Xception	97.79

TABLE 20. The Ranked models based on the average accuracies of the six DL models.

Models	Avg (%)	Rank
SAFF-ConvNeXt	99.75	1
EfficientNet-B0	98.78	2
VGG19	98.55	3
Xception	97.79	4
MobileNet	97.55	5
DenseNet121	97.48	6

for training and evaluating DL models, developing diagnostic tools, supporting clinical research, and teaching students about kidney stone disease.

From Table 21, the comparative evaluation of the six DL models—SAFF-ConvNeXt, DenseNet121, EfficientNet-B0 , MobileNet, VGG19, and Xception—revealed notable variations in classification performance across all metrics. SAFF-ConvNeXt achieved the highest overall performance, attaining an accuracy of 95.26%, specificity of 95.14%, and recall of 95.14%, which collectively indicated a strong capability in correctly identifying both positive and negative cases. Its F1-score (95.23%) further confirmed a balanced precision–recall trade-off, highlighting its robustness and reliability.

MobileNet followed closely, with an accuracy of 92.09% and an F1-score of 91.98%, showing that it maintained strong precision (92.84%) and recall (91.71%) values while being more lightweight in architecture compared to ConvNeXtBase. This suggested that MobileNet offered a competitive alternative with slightly reduced computational complexity.

Xception also demonstrated solid results, with an accuracy of 89.72% and F1-score of 89.72%, indicating consistent predictive capability. However, its slightly lower recall (89.98%) and higher false negative rate (10.02%) suggested marginally weaker sensitivity compared to ConvNeXtBase and MobileNet.

DenseNet121 achieved moderate performance, with an accuracy of 86.36% and an F1-score of 86.12%. While its results were satisfactory, the model showed a higher false

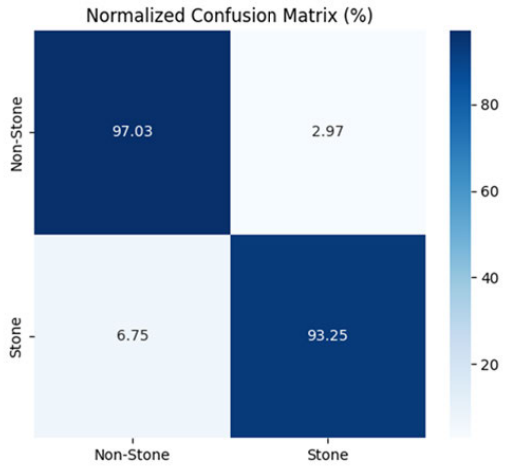


FIGURE 18. SAFF-ConvNeXt model's confusion matrix.

negative rate (14.13%), indicating that it missed more positive cases relative to the top-performing models.

In contrast, VGG19 exhibited only average classification ability, achieving an accuracy of 69.57% and an F1-score of 68.97%. Its higher false negative rate (29.28%) and lower precision (73.34%) indicated that the model struggled with misclassification and was less effective in distinguishing true positive cases.

EfficientNet-B0 produced the weakest results among the six models, with an accuracy of 53.16%, a high false negative rate of 50.00%, and a low F1-score of 34.71%, suggesting significant instability and poor generalization performance under the given dataset conditions.

Overall, SAFF-ConvNeXt emerged as the most accurate and consistent model, followed by MobileNet and Xception, while EfficientNet-B0 underperformed considerably. The results collectively indicated that deeper and more recent architectures, such as SAFF-ConvNeXt, offered superior learning capability and robustness in feature extraction compared to earlier models like VGG19 or lightweight models like EfficientNet-B0.

Based on the confusion matrix that depicted in Figure 18, The confusion matrix, once normalized, showed that the model was effective in distinguishing between the Stone and Non-Stone categories. The diagonal values revealed high classification accuracy, with 97.03% of Non-Stone samples correctly identified and 93.25% of Stone samples accurately classified. Nonetheless, there were some misclassifications: 2.97% of Non-Stone cases were wrongly predicted as Stone, and 6.75% of Stone instances were incorrectly labeled as Non-Stone. Overall, the matrix indicated that the model achieved high precision and dependability, demonstrating a strong ability to generalize across both categories.

H. DISCUSSION

The proposed SAFF-ConvNeXt model brings several benefits by combining the ConvNeXtBase backbone with SE blocks and a multi-scale feature fusion approach. This integration enhances the model's capability to extract detailed local textures as well as high-level semantic information, allowing it to identify complex anatomical differences in kidney images. The SE blocks provide adaptive channel-wise attention, enabling the network to focus on important diagnostic features while minimizing irrelevant noise, thereby boosting overall classification accuracy and interpretability. Furthermore, the multi-scale feature fusion approach aids in gathering information from various hierarchical levels, enhancing the model's robustness and ability to generalize across different imaging contexts.

I. ANALYZING RESULTS AGAINST THE STATE-OF-THE-ART

The proposed SAFF-ConvNeXt model brings several benefits by combining the ConvNeXtBase backbone with SE blocks and a multi-scale feature fusion approach. This integration enhances the model's ability to extract detailed local textures as well as high-level semantic information, allowing it to identify complex anatomical differences in kidney images. The SE blocks provide adaptive channel-wise attention, allowing the network to focus on important diagnostic features while minimizing irrelevant noise, thereby increasing overall classification accuracy and interpretability. Furthermore, the multi-scale feature fusion approach aids in gathering information from various hierarchical levels, enhancing the model's robustness and ability to generalize across different imaging contexts.

The SAFF-ConvNeXt model surpassed other contemporary methods shown in Table 22, except for Islam et al. [16], demonstrating its efficiency in the early identification of CKDs for healthcare professionals. The method outlined by Islam et al. [16] employed augmentation techniques to tackle class imbalance by increasing the number of samples in the minority class. However, if an increase is made prior to dividing the data into training and testing sets, identical data points may appear in both sets. This can lead the model to overestimate its performance in the testing set, resulting in overly optimistic results. The issue of applying oversampling before the split can result in data leakage, causing overfitting and biased results. In this research, undersampling was only applied to training data after division into training and testing sets to avoid data leakage. This strategy guaranties that the model can generalize well to new and unseen data, producing a more accurate assessment of its performance. Nevertheless, this approach may lead to slightly lower performance compared to other studies that utilize oversampling before the split, as these methods may produce inflated performance estimates due to data leakage.

The model's ability to identify subtle changes in imaging data that serve as early indicators of the disease allows for timely intervention, enabling clinicians to act before the condition worsens. The high accuracy, adaptability, and pattern recognition capabilities of the SAFF-ConvNeXt model facilitate earlier diagnoses, which are crucial to slowing disease progression, improving patient outcomes, and reducing time and costs related to the diagnostic process.

TABLE 21. Performance comparison of the six DL models based on the test set of the Axial CT Imaging Dataset.

Model	Accuracy (%)	Specificity (%)	FNR (%)	NPV (%)	Precision (%)	Recall (%)	F1-score (%)
SAFF-ConvNeXt	95.26	95.14	4.86	95.37	95.37	95.14	95.23
DenseNet121	86.36	85.87	14.13	87.24	87.24	85.87	86.12
EfficientNet-B0	53.16	50.00	50.00	26.58	26.58	50.00	34.71
MobileNet	92.09	91.71	8.29	92.84	92.84	91.71	91.98
VGG19	69.57	70.72	29.28	73.34	73.34	70.72	68.97
Xception	89.72	89.98	10.02	89.86	89.86	89.98	89.72

TABLE 22. A comparison between the proposed model and the results achieved by state-of-the-art.

References	Methodology	Performance in accuracy	Dataset
Hossain, T. et al. [18]	DT, KNN, and SVM, RF, and ALBP	99%	CTK
Islam, M. N. et al. [16]	SWIN, CCT, EANet, Inceptionv3, ResNet, and VGG16.	99.30% by SWIN	CTK
Hossain, M. S. et al. [19]	CNN	99.9 %	CTK
Afroj, M. et al. [20]	CNN	97.98%	CTK
Majid, M. et al. [21]	GLCM, RF, SVM, GB, LGBM, ResNet-101 and DenseNet-121	94.09 %	CTK
Pande, S. D., and Agarwal, R. [22]	YOLOv8	82.52%	CTK
Islam, M. N. et al. [23]	CNN and spatial attention	99.52%	CTK
Yildirim et al. [24]	CNN	96.82%	CTK
Qadir, A. M., Abd, D. F. [25]	DenseNet-201	99.44%	CTK
Proposed Model	**SAFF-ConvNeXt**	**99.754%**	**CTK**

This innovation not only reduces the burden on radiologists but also equips them with an automated system to evaluate kidney function precisely, thereby decreasing the likelihood of misdiagnosis.

In summary, the SAFF-ConvNeXt model's outstanding performance in classifying CKDs, with an accuracy rate of 99.754%, demonstrates its potential to enhance early detection and improve patient outcomes while also supporting healthcare providers in their diagnostic efforts.

V. CONCLUSION

This research introduced the SAFF-ConvNeXt, an innovative and enhanced DL model designed for the identification of chronic kidney diseases (CKDs). The proposed model combined the ConvNeXtBase backbone with SE blocks and a multi-scale feature fusion approach. It improved the early detection of CKDs for healthcare providers by its capability to extract detailed local textures as well as high-level semantic information, allowing it to identify complex anatomical differences in kidney images. The research used the CTK dataset, which comprises four different categories: cysts, stones, normal, and tumors, to train and evaluate the DL model. The initial steps for preparing the data involved normalizing and resizing the images. The CTK dataset was subsequently divided, with 85% designated for training (10,579 CT scans) and 15% for testing (1,867 CT scans). Testing performed on the CTK dataset showed that the SAFF-ConvNeXt model surpassed five other classifiers: DenseNet121, EfficientNet-B0, MobileNet, VGG19, and Xception. It achieved remarkable results in terms of 99.75% accuracy, 99.83% specificity, 99.38% sensitivity, 99.42% precision, and 99.39% F1-score.

In future research, the SAFF-ConvNeXt model could be adapted to process multi-modal medical data, such as integrating CT, MRI, and ultrasound images, which may lead to improved diagnostic precision in more challenging or unclear cases. To facilitate easier deployment, upcoming studies could concentrate on model compression and acceleration strategies, including techniques like pruning, quantization, or knowledge distillation, to minimize computational demands for real-time use on low-resource medical devices.

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